Human-in-the-Loop Predictive Safety Filtering for Minimum-Intervention Control of Recreational Surface Vehicles

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Abstract

Recreational surface vehicles should augment rather than replace human control. This paper develops an intent-preserving predictive safety architecture in which the operator supplies the desired motion and autonomy supervises only its predicted recoverability. At each control cycle, a nonlinear underactuated vessel model evaluates whether applying the admissible human command for one interval leaves a collision-free continuation within a finite escape-maneuver library. A command with such a witness passes through unchanged; otherwise, the supervisor selects the nearest witnessed command from a finite candidate set. Local LiDAR reconstruction, obstacle tracking, uncertainty inflation, and occlusion memory support prediction in environments containing static and moving obstacles without requiring a global scene map. The reported realization filters steering while retaining human throttle authority. Its escape test is therefore conditional when a witness requires subsequent propulsion changes, and does not establish an unconditional safety guarantee. We distinguish the supervisory architecture from this implementation, formulate an authority-consistent steering–propulsion extension, and establish command-preservation and sampled-minimality properties. For fixed predictions and constraints, nested expansion of candidate and backup sets can only enlarge the witnessed command set and reduce or preserve the minimum available intervention; this does not imply monotonically improving closed-loop safety. A validation protocol evaluates collision outcomes, command fidelity, and computational scalability jointly, with quantitative results reserved for future experiments. The framework provides a testable realization of a predictive safety guardian that preserves immediate human intent without optimizing a competing navigation objective.

Keywords: shared control; predictive safety filter; underactuated surface vehicle; human authority; LiDAR; sampled recoverability.

Draft status and source basis. The algorithmic description and numerical settings in this draft are based on the supplied technical report [1], available in the project as `prism-uploads/USV_Obstacle_Avoidance_and_Escape_Route_Algorithm_Report.docx`. The underlying source code, vessel-identification data, and experimental logs were not supplied. Model details not specified in that report are identified below, and the results section contains a proposed protocol and figure placeholders rather than invented findings. The supplied abstract's completed-results claims have been revised to reflect this evidence status.

1 Introduction

For a human-operated recreational vessel, choosing how and where to navigate is part of the activity rather than merely a task to automate. The operator may explore, approach an object, change

direction, or maneuver without a predefined destination. We argue that assistance in this setting should preserve the operator's admissible command whenever it retains a credible future escape and should otherwise introduce the smallest correction admitted by the safety criterion. The central question is not "Which trajectory should the vessel prefer?" but "Can the vessel execute the requested action now while retaining an escape afterward?"

This formulation separates three roles: the human specifies immediate navigation intent, a predictive supervisor assesses future recoverability, and an intervention mechanism selects a nearby admissible action when the requested action fails that assessment. The supervisor does not need to infer a destination or create a competing navigation objective. Here, *intent-preserving* has an operational meaning: preserving the measured command, or minimizing a specified distance from it. It does not imply knowledge of the operator's latent goal, awareness, or future response. Prediction of surrounding obstacles remains necessary even when inference of the operator's destination is unnecessary.

Safety filters already provide a general framework for separating performance from constraint enforcement [2, 3]. Least-restrictive reachability control [4], predictive safety certification [5, 6], and barrier-function filtering [7] all support this separation. Goal-agnostic predictive shared control with parallel trajectory sampling also predates the present work [8], as do predictive filters for assisted manual driving [9] and marine navigation [10, 11]. Consequently, neither human-first control, minimum intervention, computational parallelism, nor applying a safety filter to a ship is claimed as a new general principle.

The contribution considered here is a particular realization for human-operated underactuated surface vehicles: a local-sensing pipeline supports nonlinear, actuator-aware prediction; the exact immediate human command is tested before corrections; and acceptance depends on a separately sampled future escape continuation. Unlike a frozen-command test, the operator's request need not remain safe if held throughout the horizon. The test asks whether applying it for one control interval preserves at least one candidate continuation. This reflects the distinction between present collision freedom and future avoidability [12], while remaining a finite-horizon approximation rather than an exact viability-kernel computation.

We distinguish this *architecture* from the supplied *steering-only implementation*. The latter retains human throttle authority but searches some hypothetical escapes that change thrust. That mismatch limits the safety interpretation and motivates an explicitly identified full-command extension. Likewise, more computation can improve coverage of a fixed search problem without resolving unobserved obstacles, model errors, or missing recovery authority. These distinctions permit a constructive architectural argument without an unsupported claim of universal superiority.

Our thesis is that predictive, minimum-intervention supervision provides a natural control structure for human-operated recreational craft: human commands define the nominal motion, while autonomy acts as a *predictive safety guardian*, not a replacement navigator. Its practical value must be demonstrated along three axes simultaneously—physical collision outcomes, human-command fidelity, and computational scalability—rather than inferred from conceptual simplicity alone.

2 Related Work

2.1 Shared autonomy and minimum-intervention safety

Goal-assisting shared autonomy often models what the user is trying to accomplish. For example, Javdani et al. [13] use hindsight optimization to assist under goal uncertainty. By contrast, Reddy et al. [14] retain actions whose learned values exceed a threshold and select one close to the user input. This is a direct precedent for combining an acceptance criterion with command proximity; our acceptance criterion instead uses dynamics-based escape prediction rather than learned task value.

An especially close precedent is Broad et al. [8], who use a learned human–machine model, receding-horizon safety evaluation, and parallel sampling to minimize intervention without a specified user goal. Their work also demonstrates that exploiting greater computational resources for goal-agnostic shared control is not unique to the present architecture. The comparison therefore concerns the concrete escape test, marine dynamics, perception assumptions, and intervention metric, rather than a claimed invention of task-independent assistance.

Fisac et al. [4] develop least-restrictive safety supervision through Hamilton–Jacobi reachability. Wabersich and Zeilinger [5, 6] formulate predictive safety mechanisms that minimally modify a nominal input while seeking feasible recovery trajectories. Their assumptions and terminal constructions support safety claims that the present sampled search does not inherit. Tearle et al. [9] explicitly demonstrate predictive filtering for assisted manual driving, so replacing a learned nominal controller with a human is not, by itself, a novelty claim. More recent automotive work includes barrier-based shared control [15] and reachability-aware shared collision avoidance [16].

Control-barrier-function quadratic programs can likewise minimize deviation from a nominal command while enforcing conditions associated with forward invariance [7]. It would be inaccurate to describe all barrier methods as short-sighted geometric checks: an appropriately constructed barrier can encode future recoverability, including through backup policies [17]. The present distinction is a direct finite-library rollout test, not a categorical opposition between “predictive” and “barrier” safety. Similarly, MPC can implement safety supervision rather than autonomous navigation, and learning can be combined with hard safety filters [3].

2.2 Human–machine cooperation in maritime navigation

Huang et al. [18] propose an interpretable, interactive ship collision-avoidance system that accommodates underactuation and supports shared decision making. Their work establishes maritime human–machine cooperation as prior art. Mahipala et al. [19] use intention-aware prediction to support collaborative avoidance in mixed manned–unmanned traffic. That problem concerns cooperation across vessels; the present supervisor instead filters the immediate command on the controlled vessel. These objectives are complementary: preserving an operator’s command need not eliminate the need to predict other vessels’ behavior.

Predictive maritime safety filtering is also established. Vaaler et al. [10] formulate minimally perturbed nominal controls for a prototype ferry facing static and dynamic obstacles, while Vaaler et al. [11] combine predictive safety filtering with reinforcement learning for marine navigation. Relative to these precedents, the proposed realization emphasizes exact admissible human-command testing, a bounded escape library rather than a continuous optimal-control solve, local reconstructed

obstacle tracks, and explicit accounting for the supervisor's recovery authority. Those implementation choices require comparative validation; their combination does not establish superiority by itself.

2.3 Dynamics-aware surface-vehicle collision avoidance

Marine planning and collision avoidance span several guidance and control architectures [20]. Sun et al. [21] use a finite set of propulsion and steering commands in predictive USV collision avoidance. Abdelaal et al. [22] combine underactuated-vessel trajectory tracking, obstacle avoidance, and disturbance treatment through nonlinear MPC. Simulation-based behavior selection [23] and branching-course MPC [24] further demonstrate dynamics-aware evaluation of marine maneuvers. LiDAR-based barrier/Lyapunov control is another relevant alternative [25].

These methods establish that neither moving obstacles nor nonlinear vessel prediction is a new capability in itself. Our central design choice is to use prediction as an acceptance test around a human command, with deviation minimized only over witnessed alternatives. A fair comparison must hold sensing, control authority, model fidelity, and timing constraints fixed. The empirical question is whether this particular realization offers a useful safety–fidelity–computation trade-off, not whether all autonomous planners, MPC formulations, or learning-based controllers are inherently unsuitable for recreation.

3 Contributions and Scope

This paper develops an intent-preserving predictive supervisory architecture and documents one sampled realization. Its contributions are:

1. **A marine escape-based command filter.** Local obstacle tracking and nonlinear actuator-aware rollouts support an immediate-command/backup-continuation test. The exact admissible human steering passes through whenever a sampled escape exists; otherwise, a nearest witnessed candidate is selected.
2. **Explicit selection and refinement properties.** We establish command preservation, minimum deviation over the accepted finite set, and a monotonic refinement property for nested candidate and escape libraries under fixed predictions. These are search properties, not unconditional collision-avoidance guarantees.
3. **Separation of architecture and recovery authority.** The steering-only realization is distinguished from an authority-consistent extension that can modify steering and propulsion using a normalized full-command distance. The latter is a proposed design, not an implemented result.
4. **A joint validation framework.** The evaluation separates frozen-state search quality from closed-loop collision outcomes and measures command fidelity, computational deadlines, sensing degradation, and dependence on hypothetical throttle actions.

The intended advance is this concrete formulation and its testable trade-offs, not a claim to be the unique or ultimate architecture. The term *human-in-the-loop* describes the interface; participant studies and occupied-craft validation remain to be completed. Numerical settings below describe the supplied implementation unless explicitly labeled as an extension.

4 Problem Formulation and Modeling

4.1 Supervisory architecture and decision contract

Let $a_{h,k}$ be an admissible human command and let $\widehat{\mathcal{U}}_{\mathrm{esc},k}$ denote the set of commands with an accepted escape witness under a specified prediction model, horizon, uncertainty treatment, and recovery-authority contract. For a nonempty accepted set, the architecture is

$$a_k^* = \begin{cases} a_{h,k}, & a_{h,k} \in \widehat{\mathcal{U}}_{\mathrm{esc},k}, \\ \arg\min\limits_{a \in \widehat{\mathcal{U}}_{\mathrm{esc},k}} d(a, a_{h,k}), & a_{h,k} \notin \widehat{\mathcal{U}}_{\mathrm{esc},k}. \end{cases} \tag{1}$$

The distance is nonnegative and vanishes only for identical commands. Thus witnessed admissibility has priority over command fidelity; when the human command is admitted, zero intervention is optimal. If the accepted set is empty, the supervisor must report this explicitly and invoke a separately defined fallback. No command becomes safe merely because it is the best remaining fallback.

This contract specifies what the supervisor selects, not how its accepted set is computed. A finite maneuver search, predictive optimizer, reachability construction, or suitable barrier formulation can instantiate related contracts [2, 3]. The present paper studies a sampled escape-search implementation. Its meaning of "necessary intervention" is relative to that implemented criterion, not to an omniscient assessment of every physically possible motion.

4.2 Command interface and control authority

Let t_k denote a control update, with interval $\Delta_c = 0.10\,\mathrm{s}$. The human requests normalized throttle $h_{T,k} \in [-1, 1]$ and steering $h_{\delta,k}$. Positive and negative throttle denote ahead and reverse propulsion. Define the admissible steering request

$$\bar{\delta}_{h,k} = \mathrm{clip}(h_{\delta,k}, -\delta_{\max}, \delta_{\max}), \qquad \delta_{\max} = 30^\circ. \tag{2}$$

The applied command is

$$a_k^* = (h_{T,k}, \delta_k^*). \tag{3}$$

Consequently, unchanged passage refers to an already admissible request, not to an out-of-range raw joystick value. Throttle is not an optimization variable of the implemented filter. Minimum intervention is measured in steering-command space, not in trajectory deviation, energy, or the operator's unobserved long-term goal.

4.3 Nonlinear vessel and actuator model

The reported predictor uses the state

$$x = [n, e, \psi, u, v, r, T, \delta]^\top, \tag{4}$$

where (n, e) is position in a local north–east frame, ψ is heading, (u, v) are body-frame surge and sway velocities, r is yaw rate, and (T, δ) are actual thrust and steering states. Set $\eta = [n, e, \psi]^\top$ and

$\nu = [u, v, r]^\top$. A standard horizontal-plane representation is [26]

$$\dot{\eta} = J(\psi)\nu, \qquad J(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{5}$$

$$M\dot{\nu} + C(\nu)\nu + d(\nu) = \tau(T, \delta) + w. \tag{6}$$

Here M includes inertia and added-mass effects, C contains inertial coupling, d represents hydrodynamic damping, and τ maps the two actuator states to generalized forces and yaw moment. The vehicle is underactuated because these two commands cannot independently assign all three planar generalized forces. The report describes surge drag, sway and yaw damping, and cross-flow strip drag in its predictor [1].

For mathematical exposition, first-order actuator models with steering-rate saturation can be written as

$$\dot{T} = \frac{T_{\mathrm{cmd}}(h_T) - T}{\tau_T}, \tag{7}$$

$$\dot{\delta} = \mathrm{clip}\left(\frac{\delta_{\mathrm{cmd}} - \delta}{\tau_\delta}, -\dot{\delta}_{\max}, \dot{\delta}_{\max}\right), \quad |\delta_{\mathrm{cmd}}| \le \delta_{\max}. \tag{8}$$

Equations (6)–(8) specify a representative model structure consistent with the report, not a verified transcription of unavailable source code. The exact thrust map, inertia, damping laws, actuator constants, hull dimensions, and rate limits must be supplied before numerical reproduction. The disturbance w identifies environmental/model mismatch; no validated bound on it is available here. Nominal prediction sets $w = 0$ and therefore does not provide robust constraint satisfaction against arbitrary wind, waves, or current.

Writing the nominal dynamics as $\dot{x} = f(x, a)$, trajectories are integrated using classical fourth-order Runge–Kutta. The reported grid is

$$s_0 = 0, \quad s_1 = 0.10, \quad s_{\ell+1} = 0.10 + 0.20\ell \quad (\ell = 1, \ldots, 50), \tag{9}$$

in seconds relative to t_k. Thus the nominal 10 s horizon produces an actual final prediction time $H_p = 10.1$ s. This distinction should be retained in plots and runtime accounting. Numerical integration accuracy is not itself a collision-avoidance guarantee.

4.4 Local LiDAR reconstruction and tracking

The simulated sensor returns the first intersection on each of 360 angular rays within 65 m. Given range ρ_ℓ, body-frame bearing θ_ℓ, and the vessel pose, a valid return is transformed into the local frame as

$$p_\ell = \begin{bmatrix} n \\ e \end{bmatrix} + \rho_\ell \begin{bmatrix} \cos(\psi + \theta_\ell) \\ \sin(\psi + \theta_\ell) \end{bmatrix}. \tag{10}$$

The filter receives these observations, not obstacle ground-truth identities or trajectories. The report's instantaneous-scan abstraction does not account for rolling-scan motion distortion. Practical use would require synchronized pose estimation and calibration; onboard LiDAR alone does not supply every vessel-state component.

Adjacent returns are grouped when their separation is no more than 1.5 m, including the angular wraparound seam. For at least three points, an algebraic circle fit is accepted when the fit has rank

three, its radius lies between 0.2 and 12 m, and the largest radial residual is below 0.12 m. Rejected fits are replaced by a centroid-centered patch whose radius is the maximum observed point distance plus 0.6 m; groups with fewer than three points use the reported default radius of 0.6 m. These are buffered representations of visible returns, not certified enclosures of an object's unseen geometry.

A track has state

$$z_o = (p_o, R_o, \hat{v}_o, a_o, \sigma_{v,o}), \tag{11}$$

where $p_o \in \mathbb{R}^2$ is its current estimated center, R_o its radius, $\hat{v}_o$ its velocity, a_o the time since observation, and $\sigma_{v,o}$ a velocity-uncertainty allowance. Each detection is associated to the nearest unused predicted track using

$$d_{\text{assoc}} = \|p_{\text{new}} - (p_{\text{old}} + \hat{v}_{\text{old}}\Delta_s)\|_2 + |R_{\text{new}} - R_{\text{old}}| < 2.0\,\text{m}, \tag{12}$$

with scan interval Δ_s. Matched velocities use the reported update

$$\hat{v}_o^+ = 0.5\hat{v}_o^- + 0.5\frac{p_{\text{new}} - p_{\text{old}}}{\Delta_s}. \tag{13}$$

The uncertainty allowance has a floor of 0.12 m/s. Unmatched tracks are propagated for up to 3 s. Track memory helps retain previously observed objects during occlusion, but cannot reveal an object that has never been detected. The scalar allowance is not a calibrated probability distribution, and nearest-neighbor association does not eliminate identity switches.

4.5 Predicted obstacles and hull clearance

At future time $s \geq 0$, the tracked center and inflated radius are

$$\hat{p}_o(s) = p_o + \hat{v}_o s, \tag{14}$$

$$\hat{R}_o(s) = R_o + \sigma_{v,o}(s + a_o) + \tfrac{1}{2}a_{\max}s^2\chi_o, \tag{15}$$

$$\chi_o = \mathbf{1}\{\|\hat{v}_o\|_2 > 0.1\,\text{m/s or } \sigma_{v,o} > 0.3\,\text{m/s}\}, \tag{16}$$

where $a_{\max} = 0.025\,\text{m/s}^2$. The current center of an unseen track is already propagated to t_k; the age term enlarges its uncertainty rather than advancing its center twice. These settings are engineering allowances from the report, not verified bounds covering every obstacle maneuver. In particular, an object classified as stationary can subsequently accelerate.

Represent the hull by an oriented rectangle of length L and beam B. Define the obstacle center in hull coordinates and its offset from the rectangle half-extents as

$$q_o = R_2(\psi)^\top\left(\hat{p}_o(s) - \begin{bmatrix} n \\ e \end{bmatrix}\right), \qquad d_o = |q_o| - \begin{bmatrix} L/2 \\ B/2 \end{bmatrix}, \tag{17}$$

where R_2 is the planar rotation matrix and absolute value acts componentwise. The signed rectangle-distance expression gives

$$g_o^{\text{geom}}(x, s) = \|\max(d_o, 0)\|_2 + \min\{\max(d_{o,1}, d_{o,2}), 0\} - \hat{R}_o(s). \tag{18}$$

A positive value indicates separation of the hull and inflated circle. At a propagation sample with step length Δ, the report further subtracts the sweep allowance

$$m_{o,\text{sweep}} = \frac{\Delta}{2}\left(\sqrt{u^2 + v^2} + |r|\frac{\sqrt{L^2 + B^2}}{2} + v_{o,\max}\right), \tag{19}$$

where $v_{o,\max}$ denotes the obstacle-speed allowance used by the collision routine. Its exact construction must be specified with the source code. The final checked clearance is

$$G(x, s; \Delta) = \min_{o} \left\{ g_o^{\text{geom}}(x, s) - m_{\text{safe}} - m_{o,\text{sweep}} \right\}, \quad m_{\text{safe}} = 0.7\,\text{m}. \tag{20}$$

The initial-state check omits the sweep term. With no tracks, the obstacle minimum is $+\infty$, which is absence of a detected constraint, not evidence that unobserved space is empty. A continuous-time clearance guarantee would require valid interval-wide motion bounds and adequate geometry/error enclosures. The sample-based sweep allowance alone establishes neither.

5 Predictive Minimum-Intervention Method

5.1 Candidate-now, backup-later prediction

For a candidate steering c, every predicted branch first applies the same immediate command. At future time s, its input is

$$a_{c,b}(s) = \begin{cases} (h_{T,k}, c), & 0 \le s < \Delta_c, \\ \pi_b(x(s), s; c, h_{T,k}), & \Delta_c \le s \le H_p. \end{cases} \tag{21}$$

Here b indexes a backup policy from $\mathcal{B}(c)$. The steering choices are $\{-30°, -15°, 0°, 15°, 30°, c\}$, crossed with the four modes in Table 1. This gives 24 branches, including possible duplicates when c equals a fixed library angle. The turn-then-straighten switch is at prediction time $s > 2\,\text{s}$, as described in the report.

Table 1: Reported backup modes. These are hypothetical continuations, not automatically executed emergency actions.

Mode	Throttle after the first interval	Steering after the first interval
Hold throttle	Current human throttle $h_{T,k}$	Fixed sampled angle
Full ahead	Maximum ahead command	Fixed sampled angle
Braking	Reverse for $u > 0.15\,\text{m/s}$; ahead for $u < -0.15\,\text{m/s}$; neutral otherwise	Fixed sampled angle
Turn then straighten	Current human throttle $h_{T,k}$	Sampled angle until $s > 2\,\text{s}$, then zero

Let $x_{c,b}(s_\ell)$ denote the numerical prediction. The sampled escape value is

$$\widehat{V}_k(c) = \max_{b \in \mathcal{B}(c)} \min_{\ell = 0, \ldots, 51} G(x_{c,b}(s_\ell), s_\ell; \Delta_\ell), \tag{22}$$

where $\Delta_0 = 0$ and $\Delta_\ell = s_\ell - s_{\ell-1}$ for $\ell \ge 1$. A candidate has a *sampled escape witness* if $\widehat{V}_k(c) > 0$. A branch is permanently rejected after its first nonpositive checked clearance, even if subsequent samples become separated again. Among successful branches, the witness with the largest minimum clearance may be retained for visualization. Its displayed path is a conditional prediction, not the vessel's committed future trajectory.

Equation (22) is a finite-grid, finite-library test. For fixed numerical dynamics, predictions, and constraints, adding branches cannot decrease $\widehat{V}_k(c)$. This does not make the accepted set a certified inner approximation of the true continuous-time viability kernel: discretization, model error, and incomplete perception can also produce false acceptance. Conversely, a failed test can reject a state for which an unsampled escape exists.

5.2 Human-first acceptance and nearest correction

The filter first evaluates $\widehat{V}_k(\bar{\delta}_{h,k})$. If positive, it returns the exact admissible request with status $s_k = 0$. Only if this test fails does it form

$$\mathcal{C}_k = \{\bar{\delta}_{h,k}\} \cup \{-30^\circ, -28^\circ, \ldots, 28^\circ, 30^\circ\}, \tag{23}$$

$$\mathcal{A}_k = \{c \in \mathcal{C}_k : \widehat{V}_k(c) > 0\}. \tag{24}$$

For a nonempty accepted set, the rule is

$$\delta_k^* = \begin{cases} \bar{\delta}_{h,k}, & \widehat{V}_k(\bar{\delta}_{h,k}) > 0, \\ \arg\min\limits_{c \in \mathcal{A}_k} |c - \bar{\delta}_{h,k}|, & \widehat{V}_k(\bar{\delta}_{h,k}) \le 0,\ \mathcal{A}_k \neq \varnothing. \end{cases} \tag{25}$$

The second case returns status $s_k = 1$. Sorting candidates by absolute steering deviation and selecting the first accepted candidate implements this minimization. Ties use the stable candidate ordering of the implementation; a reproducible release must document that ordering. The correction is minimal only among accepted sampled commands. No universal 1° error bound relative to a continuous optimum follows from the 2° lattice, since the accepted steering set can be disconnected or narrower than the grid spacing.

5.3 No-witness fallback

When $\mathcal{A}_k = \varnothing$, the filter raises status $s_k = 2$ and selects a best-effort command. Let $t_{\mathrm{surv}}(c, b)$ be the duration credited to the branch before its first failed sampled check, and $g_{\min}(c, b)$ its minimum checked gap. The report uses

$$Q_k(c, b) = t_{\mathrm{surv}}(c, b) + \epsilon_g \,\mathrm{clip}\big(g_{\min}(c, b), -100\,\mathrm{m}, 100\,\mathrm{m}\big), \tag{26}$$

$$\delta_k^* = \arg\max_{c \in \mathcal{C}_k} \max_{b \in \mathcal{B}(c)} Q_k(c, b), \quad \epsilon_g = 0.001\,\mathrm{s/m}. \tag{27}$$

This weighted score emphasizes survival duration but is not strictly lexicographic: the gap term can change the ordering of close survival times. Status 2 means *no escape found in the tested library*, not *collision proven inevitable*. Nor does the selected fallback qualify as safe. A deployed interface should clearly distinguish this condition from successful assistance; an emergency-stop guarantee is unavailable when throttle cannot be overridden.

5.4 Algorithm and computational budget

Algorithm 1: sampled predictive steering supervision

1. Obtain the current vessel state and admissible human command. Reconstruct LiDAR detections, update associations and velocities, and retain eligible unseen tracks.
2. Predict the 24 branches for the exact human steering using Equations (21)–(22). If any survives, apply $(h_{T,k}, \bar{\delta}_{h,k})$ and return status 0.
3. Form $\mathcal{C}_k$ and sort by $|c - \bar{\delta}_{h,k}|$. Evaluate candidate escape values. If any candidate survives, apply the first accepted steering with the unchanged human throttle and return status 1.

4. If none survives, use Equation (27), keep human throttle unchanged, and return status 2.
5. Record the command, status, selected witness, checked clearance, and execution time; repeat at the next control update.

The correction stage contains at most 32 candidates and $32 \times 24 = 768$ branches. If the failed initial human-command test is recomputed within that stage, the full call evaluates up to $24 + 768 = 792$ branches. Each full branch has 51 propagation steps and four derivative evaluations per RK4 step. With N_o tracks, the principal geometry workload scales as $O(|\mathcal{C}_k|\,|\mathcal{B}|\,51\,N_o)$. The report describes vectorized propagation; sorting determines selection priority but does not necessarily imply sequential early termination of the batch computation. A bounded branch count supports predictable workload sizing, not a measured real-time claim. End-to-end latency must be compared with the 100 ms control deadline on the target hardware.

5.5 Properties and limits of the safety statement

Proposition 1 (Command preservation and sampled minimality)**.** *For fixed state, tracks, human throttle, candidate set, and deterministic escape-test outputs, Equation* (25) *preserves every admissible human steering command having a sampled escape witness. If the human command fails and $\mathcal{A}_k$ is nonempty, the returned steering has minimum absolute deviation among all members of $\mathcal{A}_k$.*

Proof. The first property follows from the first branch of Equation (25). In the second case, candidates are traversed in nondecreasing absolute deviation. Any candidate with strictly smaller deviation than the first accepted one has already failed the same test. Therefore no accepted candidate has a smaller deviation. Equal-deviation candidates are resolved by the stated tie rule. □

This proposition is an algorithmic selection property, not a physical safety theorem. Formal predictive or backup-based safety methods require additional assumptions supporting the continued availability and execution of recovery control [6, 17]. The present test has no terminal invariant-set condition, no verified model-error enclosure, and no enforceable commitment to its backup trajectory. Repeating a finite-horizon test does not, by itself, establish recursive feasibility.

Authority-consistent recovery. The library permits future thrust changes that Equation (3) cannot enforce. A braking witness can therefore make the immediate human command acceptable even when a persistently accelerating operator would prevent that recovery. For a future design, one option is an explicitly authorized emergency thrust override. Another is a steering-only robust test against a defined set of possible future throttle signals, whose logical form is

$$\exists\ \pi_\delta \text{ causal} \quad \forall\ h_T(\cdot) \in \mathcal{H}_T : \quad G(x(s), s) > 0 \text{ over the required interval.} \tag{28}$$

Here the shorthand $G(x(s), s)$ denotes the required continuous-time clearance criterion, with suitable uncertainty treatment; it is not just the sampled check in Equation (20). Equation (28) is a proposed design requirement, not a property of the reported 24-branch search. Even retaining only constant-human-throttle backups assumes that the operator will hold that throttle. Arbitrary future throttle can make collision avoidance impossible with steering alone. A credible safety specification must state who can execute recovery, with what delay, and under which assumptions.

5.6 Authority-consistent full-command extension

A proposed extension grants the supervisor authority over both steering and propulsion when the human command fails the escape test. This changes the applied-control policy and is not part of the supplied steering-only implementation. Using normalized throttle rather than the physical thrust state, write $a = (h_T, \delta_c)$ and $a_{h,k} = (h_{T,k}, \bar{\delta}_{h,k})$. One interpretable distance is

$$d_W(a, a_{h,k}) = \left[w_T \left(\frac{h_T - h_{T,k}}{s_T} \right)^2 + w_\delta \left(\frac{\delta_c - \bar{\delta}_{h,k}}{s_\delta} \right)^2 \right]^{1/2}, \tag{29}$$

where $w_T, w_\delta > 0$ are declared preference weights and $s_T, s_\delta > 0$ normalize command units and scales. Weights determine what "closest" means across propulsion and steering; they do not trade collision constraints against preference penalties.

A finite full-command set $\mathcal{C}_k^{\text{full}}$ includes $a_{h,k}$ exactly. Each candidate is applied hypothetically for the first interval, and every backup must lie within the supervisor's actual actuator authority, including reversal restrictions, rate limits, and switching delays. Defining

$$\widehat{\mathcal{U}}_{\text{esc},k}^{\text{full}} = \{a \in \mathcal{C}_k^{\text{full}} : \widehat{V}_k^{\text{auth}}(a) > 0\}, \tag{30}$$

Equation (1) with $d = d_W$ provides exact full-command pass-through or minimum weighted correction. The filter must also be able to execute the stored recovery policy when needed; simply possessing thrust authority is insufficient if the recovery is continually deferred. The latest perception must revalidate any retained witness. Terminal safety conditions, credible uncertainty bounds, and a verified switching/deadline policy remain necessary for a formal guarantee. This extension removes the logical mismatch between hypothetical and available inputs, but does not by itself solve the other assurance requirements.

5.7 Computational refinement: what more search can establish

Consider a frozen control snapshot. Fix the vessel state, human request, dynamics, perception, horizon, time grid, margins, authority contract, and deterministic branch scores. Let refinement levels $j = 1, 2, \ldots$ satisfy

$$\mathcal{C}^{(j)} \subseteq \mathcal{C}^{(j+1)}, \qquad \mathcal{B}^{(j)}(c) \subseteq \mathcal{B}^{(j+1)}(c) \quad \text{for every } c \in \mathcal{C}^{(j)}. \tag{31}$$

Every set retains the exact human request. Old branches retain their numerical evaluations. Write $\widehat{V}^{(j)}(c)$ for Equation (22) at level j, and $\mathcal{A}^{(j)} = \{c \in \mathcal{C}^{(j)} : \widehat{V}^{(j)}(c) > 0\}$.

Proposition 2 (Nested-search refinement)**.** *Under Equation* (31) *and the fixed-snapshot assumptions,*

$$\widehat{V}^{(j+1)}(c) \geq \widehat{V}^{(j)}(c), \qquad \mathcal{A}^{(j)} \subseteq \mathcal{A}^{(j+1)}. \tag{32}$$

If $D^{(j)} = \min_{c \in \mathcal{A}^{(j)}} |c - \bar{\delta}_h|$, with $D^{(j)} = +\infty$ for an empty set, then $D^{(j+1)} \leq D^{(j)}$. The same statement holds for nested full-command sets and a fixed distance d_W.

Proof. For each old candidate, the maximum over an expanded backup library includes every previously evaluated branch, so its escape value cannot decrease. Every old accepted candidate remains accepted, and the larger candidate set can add further accepted commands. Minimizing the same distance over a superset cannot increase the optimum. If no candidate has yet been accepted, the extended-real convention covers that case. □

This result formalizes a limited but useful benefit of computation: more nested control alternatives cannot worsen the best available command match for the *same* prediction problem. It does not establish greater clearance for the selected command, lower collision incidence on new closed-loop trajectories, or convergence to a continuous optimum. A newly discovered, closer command may have a smaller positive clearance margin. Additional latency can also reduce practical safety. Mere trajectory counts do not establish nesting; a finer but shifted grid may discard old samples.

Increasing the horizon or adding uncertainty hypotheses changes the constraint test rather than only its search coverage. For example, with a fixed obstacle-hypothesis set Ω, a conservative finite-hypothesis extension would test

$$\widehat{V}_{\Omega}(c) = \max_{b \in \mathcal{B}(c)} \min_{\omega \in \Omega} \min_{\ell} G_{\omega}\big(x_{c,b,\omega}(s_{\ell}), s_{\ell}; \Delta_{\ell}\big). \tag{33}$$

One common backup policy must succeed for every tested hypothesis, using only information available causally. Allowing a different clairvoyant backup for each hypothesis would test a weaker condition. Enlarging Ω can decrease this value and increase intervention. Extending a trajectory's horizon adds constraints and can likewise remove a witness. Neither is covered by the nested-control monotonicity proposition, and finitely many hypotheses are not a guarantee against unmodeled motion.

Deadline-aware refinement. An incremental variant could retain the best fully evaluated witness found so far and refine while time remains. A partial rollout is not an accepted witness. At a deadline, a witnessed incumbent is minimal only among evaluated accepted commands unless every closer candidate has been conclusively rejected. Failure to find a witness before exhausting the search budget is not proof that no witness exists in the intended full library; it must be logged separately from exhaustive status 2. A previously stored witness must be updated and rechecked before reuse. These rules define a proposed budgeted implementation, not an anytime guarantee of the reported vectorized code.

6 Evaluation Protocol and Results Placeholders

No simulation traces, participant data, or vessel experiments are available in the supplied materials. This section defines the proposed evaluation and reserves space for its results. All scenario comparisons, ablations, and human-factors measurements below are planned rather than completed.

6.1 Reproducible setup

Table 2 lists the settings documented in the report. Before running the study, complete the missing plant and actuator parameters identified in Section 4, and disclose the state estimator, scenario geometry, noise process, scan timing, numerical tolerances, random seeds, trial count, software version, and computing hardware. The plant simulation should be independent of the prediction routine or deliberately perturbed; identical plant and predictor models would conceal an important source of risk. Ground-truth obstacle states may be used for scoring but must not be provided to the filter.

Table 2: Documented configuration; these are algorithm settings, not measured performance.

Parameter	Reported value
Control interval / backup integration step	0.10 s / 0.20 s
Nominal horizon / implemented final prediction time	10.0 s / 10.1 s
Steering bounds / correction lattice spacing	$\pm 30^\circ$ / 2°
Backup branches per candidate	24
LiDAR range / beam count	65 m / 360
Fixed clearance buffer	0.7 m
Track-memory duration	3.0 s
Velocity-uncertainty floor	0.12 m/s
Moving-track acceleration allowance	$0.025\,\mathrm{m/s^2}$

6.2 Scenarios, comparators, and ablations

The scenario suite should include (i) open water with benign steering; (ii) an isolated obstacle and a narrow passage; (iii) crossing, head-on, and overtaking moving obstacles; (iv) multiple obstacles with competing escape directions; (v) temporary occlusion, sparse returns, and newly revealed objects; and (vi) delayed detection, sustained high throttle, and dead ends beyond the horizon. Geometric encounter labels are used for scenario organization, not as a claim of implemented navigational-rule compliance.

Use matched, prerecorded command sequences to compare controllers without changes in operator behavior obscuring algorithmic differences. Include safe commands, delayed avoidance, commands aimed toward obstacles, rapid steering reversals, and persistent throttle during predicted braking escapes. These scripted tests assess the filter, not human experience. A separate counterbalanced participant study is needed for claims about perceived authority, trust, workload, or comfort; participant recruitment, consent, training, and any required ethics approval remain to be specified.

The principal comparators should be unfiltered human commands; a frozen-command predictor that tests holding the candidate throughout the horizon; and a steering filter that selects the largest-clearance accepted candidate instead of the nearest one. The last comparison isolates the effect of the minimum-deviation objective while holding perception and the branch library fixed. A predictive-optimization or high-order-CBF comparator is also useful, but must use matched actuation authority, sensing, and dynamics; published guarantees or runtimes cannot simply be transferred to a different implementation [7, 25].

Component ablations should remove track memory, radius inflation, and the sweep allowance separately, then vary the prediction horizon, grid resolution, and maneuver library. The most important authority ablation compares the full hypothetical library with a constant-human-throttle library and, separately, a controller explicitly allowed to execute emergency thrust changes. Those variants have different assumptions and must not be presented as interchangeable safety guarantees.

6.3 Metrics and evaluation questions

For K recorded control updates, define $d_k = |\delta_k^* - \bar{\delta}_{h,k}|$ and a preregistered numerical equality tolerance ε_δ. Report

$$P_{\text{pass}} = \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}\{d_k \le \varepsilon_\delta\}, \tag{34}$$

$$I_{\text{rate}} = 1 - P_{\text{pass}}, \qquad D_{\text{mean}} = \frac{1}{K} \sum_{k=1}^{K} d_k, \tag{35}$$

$$D_{\text{peak}} = \max_k d_k, \qquad R_{\text{no witness}} = \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}\{s_k = 2\}. \tag{36}$$

These are unconditional measures: pass-through rate is not a safety score, and intervention rate includes unsafe fallback commands. Separately verify that every status-0 output equals the admissible human request and that every status-1 output minimizes deviation over the accepted lattice. A full candidate sweep on logged snapshots provides an offline check of the latter property.

Safety-related outcomes should include per-trial collision incidence, true hull–obstacle minimum clearance, minimum checked clearance, time of first intervention, and duration in status 2. Evaluate physical collisions using an independent finer-time collision check, not just the controller's own inflated samples. Record whether each accepted witness changes throttle, and report loss of recoverability when that thrust action is not executed. Computation measurements should include median, 95th and 99th percentiles, maximum observed end-to-end latency, and deadline-miss fraction. Report paired trial-level differences with uncertainty intervals; consecutive control samples are not independent experimental trials. No fixed sample size or statistical significance is asserted here.

6.4 Safety, command fidelity, and computational scalability

The principal empirical claim should be assessed through a safety–intervention diagram, not a single collision statistic. For each matched trial of duration T_{trial}, define accumulated normalized command deviation

$$J_{\text{int}} = \sum_{k=1}^{K} \Delta_c \, d_W(a_k^*, a_{h,k}), \qquad \bar{J}_{\text{int}} = J_{\text{int}}/T_{\text{trial}}. \tag{37}$$

For steering-only experiments, use $d_W = |\delta_k^* - \bar{\delta}_{h,k}|/\delta_{\max}$. Full-command experiments use Equation (29) with disclosed weights. Plot collision incidence against $\bar{J}_{\text{int}}$, with uncertainty intervals and computation encoded separately. Lower values on both axes are desirable, but no method is presumed to occupy that region. A measured trade-off plot is not proof of global Pareto optimality. Include minimum true clearance, no-witness events, and timeout events so that low intervention cannot conceal unsuccessful supervision. Use matched trial durations or explicitly account for early collision termination, which otherwise artificially reduces accumulated intervention.

The main comparison should include a human-nominal predictive optimization filter, motivated by Wabersich and Zeilinger [6], Tearle et al. [9], Vaaler et al. [11], and a goal-agnostic sampling-based shared controller motivated by Broad et al. [8]. These are more informative tests of the claimed advantage than only comparing against an autonomous destination planner. A suitably constructed barrier/backup filter is an additional safety-oriented comparator. Label any adapted implementation

as such, match available sensing and actuation, and report both equal-deadline comparisons and hardware/resource requirements. An exact replication of published experiments is not implied.

Two separate experiments evaluate scalability:

1. **Frozen-snapshot refinement.** Replay identical logged states and tracks with nested candidate and backup sets. Use the baseline correction-stage budget of at most 768 branches and illustrative larger caps near 1,500, 3,000, and 10,000; disclose actual unique branches and separately count the initial human test. Retain old branches, the exact human request, and identical scoring. Record accepted-set inclusion, best intervention, human-command acceptance, and runtime. The proposition predicts nondecreasing witnessed coverage and nonincreasing optimal deviation, not any particular numeric improvement. A dense reference search is a computational comparator, not a ground-truth viability oracle.
2. **Closed-loop resource scaling.** Run each configuration on matched command sequences and scenarios with its actual sensing-to-actuation delay. Report collisions, physical clearance, accumulated deviation, intervention switching, and deadline misses. Changing the horizon, uncertainty hypotheses, model fidelity, or authority policy must be studied separately from fixed-problem library refinement. Closed-loop outcomes may be nonmonotonic and all such outcomes must be retained.

A participant study should then test whether improved command fidelity corresponds to perceived authority or reduced conflict. Quantitative command agreement is an engineering outcome; perceived agency and comfort are distinct human-subject outcomes. No participant or performance result is asserted in this draft.

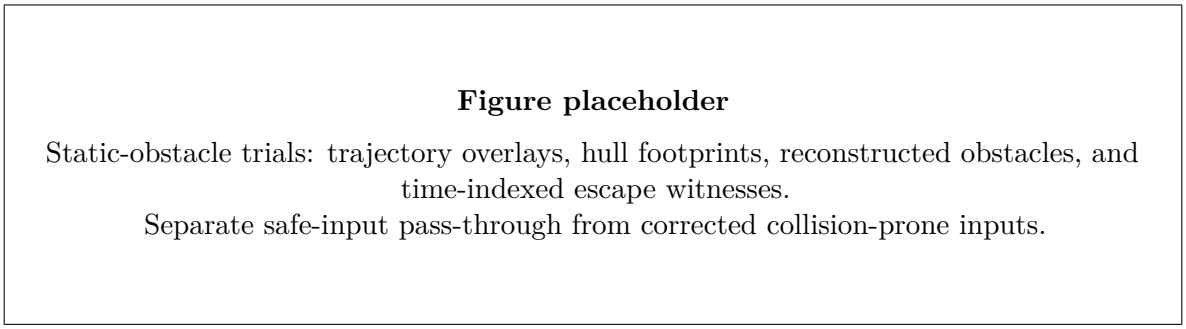


Figure 1: Planned static-scene results. Show requested-command-only and filtered trajectories under matched commands; distinguish physical obstacle geometry from inflated planning geometry. Plot witness paths as predictions, not executed trajectories. No result is shown yet.

6.5 Results reporting template

Figures 1–6 should support four distinct questions: whether admissible witnessed commands are preserved; whether corrections are minimal over the tested candidates; whether physical collision outcomes improve under matched inputs; and whether computation fits the intended deadline. Table 3 reserves the aggregate outcomes without implying measurements have been made. Claims of low intervention are meaningful only alongside clearance and collision outcomes. Likewise, a positive predicted escape value must not be reported as observed physical safety.

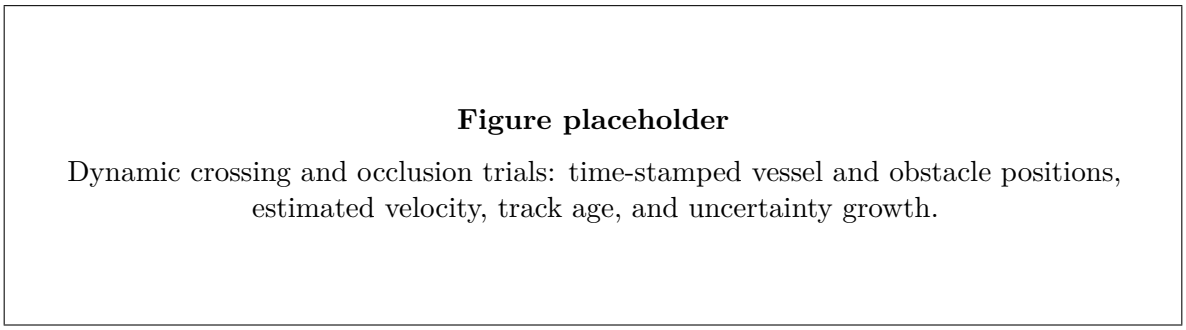


Figure 2: Planned dynamic-scene results. Include an obstacle first observed before occlusion and a separate never-observed obstacle case. Quantify estimation error and physical clearance rather than inferring robustness from trajectory appearance alone.

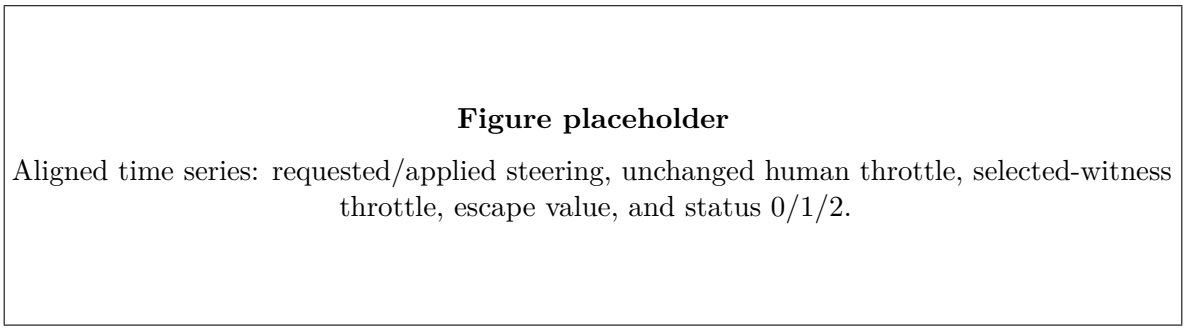


Figure 3: Planned command-preservation and authority audit. Highlight intervals where the displayed escape requires a thrust action the supervisor cannot apply. Evaluate benign pass-through, minimum-deviation corrections, and no-witness fallback separately.

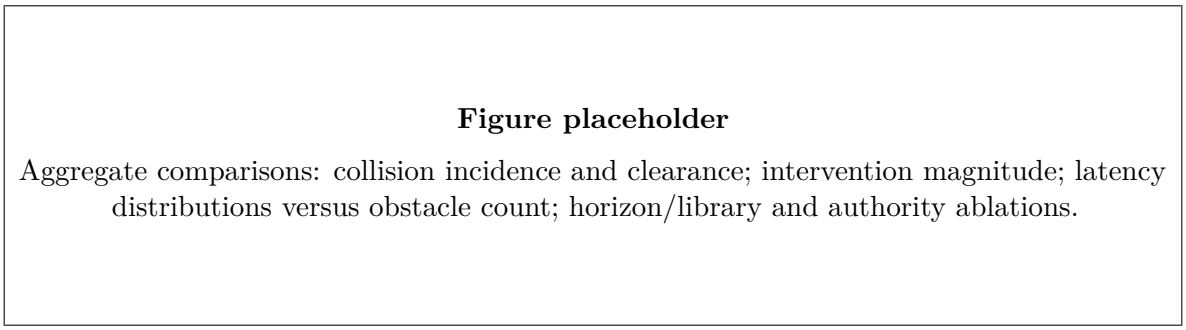


Figure 4: Planned quantitative evaluation. Provide trial counts and uncertainty intervals, include failed trials, and mark the 100 ms control deadline. Any claim of a safety–intervention trade-off must use both physical outcomes and command-deviation measurements.

Figure placeholder

Safety–intervention comparison: collision incidence versus normalized accumulated command deviation.
Separate steering-only and full-authority experiments; encode latency and show uncertainty intervals.

Figure 5: Planned joint safety and command-fidelity comparison under matched scenarios. Lower collision incidence and lower intervention are desirable, but dominance is an empirical question. Include predictive-filter and goal-agnostic shared-control comparators; no results are plotted.

Figure placeholder

Panel A: nested frozen-snapshot search coverage and minimum deviation.
Panel B: closed-loop collisions, clearance, and latency at increasing computation budgets.

Figure 6: Planned computational-scaling study. Only Panel A tests the nested-search monotonicity proposition under fixed predictions and constraints. Panel B may be nonmonotonic; mark the control deadline and distinguish library growth from longer horizons or additional uncertainty hypotheses.

Table 3: Main comparison results to be populated; additional ablations appear in the planned figures. "Pending" denotes missing experiments. Infeasibility and no-witness events must be interpreted using each controller's stated criterion.

Metric	Unfiltered	Predictive filter	Sampled shared control	Proposed
Collision incidence per trial	Pending	Pending	Pending	Pending
Minimum physical clearance	Pending	Pending	Pending	Pending
Command pass-through fraction	Pending	Pending	Pending	Pending
Normalized accumulated intervention	Pending	Pending	Pending	Pending
Mean / peak steering correction	Pending	Pending	Pending	Pending
Infeasible / no-witness fraction	N/A	Pending	Pending	Pending
99th-percentile end-to-end latency	N/A	Pending	Pending	Pending
Deadline-miss fraction	N/A	Pending	Pending	Pending

7 Discussion

7.1 Preserving human intent without claiming to infer it

The architecture expresses a deliberately limited assistance contract: the operator chooses the immediate desired action, while the supervisor tests the predicted availability of an escape. Exact command preservation avoids continuous blending when the criterion is satisfied. Nevertheless, absolute steering deviation is only a proxy for intent. A small correction can still take the vessel to the undesired side of an obstacle, and a large but brief correction may be preferable to a prolonged small one. Human preference therefore requires direct evaluation, consistent with the trade-offs observed in shared-autonomy research [13]. Neither low intervention frequency nor a visually smooth path establishes trust or perceived control.

7.2 Finite-horizon evidence versus guaranteed safety

The approach occupies a specific position between geometric reaction and formally certified safety control. It searches dynamics-consistent continuations rather than only instantaneous clearance, but its successful outcome is a sampled witness under the current model. A trajectory can remain clear until H_p and enter an inescapable configuration afterward. An unseen obstacle, poor association, or underestimated disturbance can invalidate an otherwise positive prediction. The fail/pass distinction must therefore be described as a statement about the search, not an exact classification of the inevitable-collision set [12].

There are several routes toward stronger assurance: a reachable terminal safe set, validated state and obstacle uncertainty envelopes, continuous-time or certified sampled-data clearance bounds, and explicit recovery-control authority. These requirements connect the design to predictive and backup safety methods [6, 17]. They also clarify why the current implementation should not inherit the guarantees of those methods solely because it predicts backup trajectories. Until these requirements are established, the proper description is a predictive assistance layer with conditional finite-horizon escape testing.

7.3 The throttle-authority trade-off

Leaving throttle entirely to the operator is a clear interface policy, but it limits what steering supervision can accomplish. At high speed or with insufficient turning room, no steering-only correction may avoid a collision. More subtly, the reported library can accept a command because a reverse-thrust continuation exists even though that continuation remains unenforceable. This is not resolved by increasing the number of steering samples. The design must either communicate the need for a specific operator response, restrict the acceptance criterion to a stated throttle contract, or authorize a separate emergency mode. A communicated request additionally requires a model of response delay; displaying a green escape path is not evidence that the operator can or will execute it.

7.4 Perception, environment, and recreational deployment

The local sensing pipeline avoids dependence on a complete global map, but its footprint circles and memory window do not resolve arbitrary occlusion. Beyond-range and never-observed regions remain uncertain. The documented simulation does not establish performance under spray, reflective water returns, roll and pitch, or sensor outages. These should be included in future sensor and field validation rather than absorbed into an unexplained fixed margin. Similarly, the horizontal-plane collision objective does not address grounding, capsize, excessive passenger acceleration, or every hazard relevant to occupied recreational craft.

Geometric separation also differs from navigational-rule compliance. The implementation contains no encounter classifier or explicit COLREGs constraints. This contrasts with marine methods that incorporate such behavior directly [23, 24]. A recreational implementation would need an additional, validated treatment of navigation rules and clear operator responsibilities; no compliance claim is made here. USV testing should precede claims about occupied operation, and the safety case must extend beyond collision geometry.

7.5 Architecture versus implementation

The architectural claim is that the nominal objective should remain with the human while the supervisor handles recoverability and command correction. It is not that every other computational method should be displaced. Predictive optimization, barrier constructions, reachability, and learned proposals can support related supervisory contracts [3]. Direct finite-library search is one transparent way to expose candidate commands and their predicted escapes, with incomplete coverage as its principal trade-off.

Additional computation can support nested library expansion without changing the interface or intervention objective. Section 5.7 states exactly what this buys: improved command matching for a fixed prediction problem. It cannot compensate for an invalid dynamics model, unseen hazards, or unavailable recovery inputs. This distinction separates a scalable implementation strategy from a claim of ultimate performance. Comparative evidence should identify the operating regimes in which sampling offers a favorable trade-off and those in which a predictive optimizer or another safety mechanism performs better.

7.6 Practical extensions and open questions

Adaptive sampling near accepted/rejected transitions may improve coverage but must retain old candidates to inherit the refinement result. A secondary tie-breaker favoring the previously applied steering can reduce equal-cost switching without changing the primary distance objective; broader hysteresis can sacrifice exact pass-through and must be labeled as a different policy. Neither adaptive refinement nor hysteresis is claimed as implemented here.

The most important open questions are whether the proposed realization achieves useful physical clearance with less command deviation than matched safety-filter baselines, how its performance changes under sensing and model error, and whether its required recovery inputs are actually available. The full-command extension and deadline-aware search provide concrete development paths, while human-subject evaluation must determine whether engineering command fidelity translates into a preferable recreational experience.

8 Conclusion

This paper develops intent-preserving predictive safety supervision for human-operated surface vehicles. The operator remains the source of nominal motion, while autonomy tests whether the immediate command retains a sampled escape continuation. A witnessed admissible command passes through unchanged; otherwise, the nearest witnessed command is selected, with a distinct best-effort state when the search finds no escape. This separation permits assistance without a competing autonomous navigation objective.

The reported steering-only implementation combines local obstacle reconstruction and tracking with nonlinear actuator-aware rollouts. Its selection rule preserves witnessed commands and minimizes deviation over the accepted finite set. Nested expansion of control and backup libraries improves or preserves that fixed-problem optimum, but does not imply monotonically improving closed-loop safety. A proposed full-command extension aligns hypothetical recovery actions with steering and propulsion authority; terminal, uncertainty, and timing requirements remain unresolved before a formal safety guarantee can be claimed.

Predictive minimum-intervention supervision is therefore advanced as a natural architecture to investigate, not as a universally superior or final solution. The joint evaluation of collision outcomes, command fidelity, and computational resources provides the evidence needed to assess that thesis. Quantitative and human-subject validation remain to be completed.

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